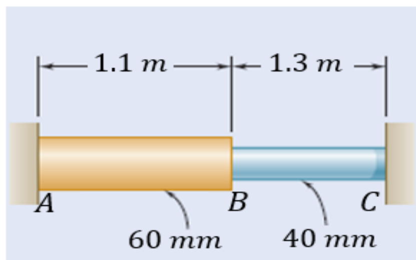
1st Practical Work: **Axial and thermal loads**

A. Thermal stresses in a compound rod made of two different materials

A rod consisting of two cylindrical portions AB and BC is restrained at both ends. Portion AB is made of brass ($E_b = 105\ GPa, \alpha_b = 20.9 \times 10^{-6}/°C$) and portion BC is made of aluminum ($E_a = 72\ GPa, \alpha_a = 23.9 \times 10^{-6}/°C$). Knowing that the rod is initially unstressed, the objective is to determine the stresses induced in AB and BC, as well as the displacement of the point B, when there is a <u>temperature rise of 42°C.</u> The length and diameter of each portion are shown in the figure.



1. Analytical solution (calculations by hand):
   a. Determine the normal stresses in the section AB & BC, σ_{AB} and σ_{BC}
   b. Determine the displacement of the point B ($\delta_B = \delta_T^{AB} - \delta_R^{AB}$)
      i. Calculate the total elongation (change of length) δ_T due to temperature that the rod would have had if it wasn't restricted. ($\delta_T = \delta_T^{AB} + \delta_T^{BC}$)
      ii. Determine (as a function of R) the change of length δ_R of the rod due to the reaction force R from the supports ($\delta_R = \delta_R^{AB} + \delta_R^{BC}$)
      iii. Knowing that $\delta_T = \delta_R$, find the value of R and deduce the resulting normal stresses σ_{AB} and σ_{BC} in the two portions of the rod.
2. Numerical solution (using the software, follow the provided guide):

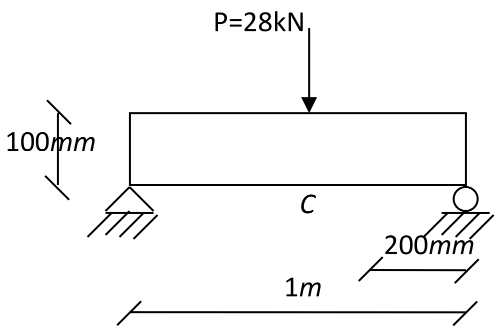
Using Abaqus (Educational Edition), launch a "Abaqus CAE":

   a. Determine the normal stresses σ_{AB} and σ_{BC} in the axial direction in the middle of the portions,
   b. and the displacement (deformation) of the point B along the axial direction

3. Compare the analytical and numerical solutions.

	1. Analytical solution:	2. Numerical solution:
Normal stress in AB section	σ_x = …….. MPa	S_{11} = …….. MPa
Normal stress in BC section	σ_x = …….. MPa	S_{11} = …….. MPa
Displacement at point B:	δ_B = …….. mm	U_1 = …. mm

B. Stress in beam



Width of the beam=25*mm*

Material Properties:

E = 200 GPa, ν = 0.3

$\sigma_{allowable}$ = 200 MPa

1. Analytical solution:
   a. Determine σ_x and τ_{xy} at point C. (C is in the middle and bottom of the beam)
   b. Determine the principal stresses σ_1 and σ_2 by drawing the state of stress at that point and using the formula or Mohr circle.
   c. Calculate the Von Mises stress at point C.

$$\sigma_v = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2}$$

   d. What is the damage at point C? (Compare Von Mises stress with $\sigma_{allowable}$)
2. Numerical solution:

Using Abaqus Education Software, make a 2D planar model of the beam above. Determine the parameters below

	3. Analytical solution:	4. Numerical solution:
Normal stress at point C:	σ_x = …….. MPa	S_{11} = …….. MPa
Shear stress at point C:	τ_{xy} = …….. MPa	S_{12} = …….. MPa
Principal stresses at point C: σ_x	σ_1 = …….. MPa (Max) σ_2 = …….. MPa (Min)	Max. Principal = …. MPa Min. Principal = …. MPa
Von Mises stress at point C	σ_V = …….. MPa	S_{Mises} = …….. MPa

3. Compare all parameters by both methods. If the results are not the same, please change your mesh size to as small as possible. Give your final comment.